# Basic Chatbot

print("My chatbot project has started!")

In [3]:
```
import random
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression

# Training data
texts = [
    "hello", "hi", "hey",
    "bye", "goodbye", "see you",
    "what is your name", "who are you",
    "what is python", "tell me about python",
    "what is machine learning", "explain machine learning",
    "thank you", "thanks"
]

labels = [
    "greeting", "greeting", "greeting",
    "goodbye", "goodbye", "goodbye",
    "name", "name",
    "python", "python",
    "ml", "ml",
    "thanks", "thanks"
]

# Convert text into numerical features
vectorizer = TfidfVectorizer()
X = vectorizer.fit_transform(texts)

# Train the ML model
model = LogisticRegression()
model.fit(X, labels)

# Responses
responses = {
    "greeting": "Hello! How can I help you?",
    "goodbye": "Goodbye! Have a nice day!",
    "name": "I am a Machine Learning chatbot.",
    "python": "Python is a popular programming language.",
    "ml": "Machine Learning helps computers learn patterns from data.",
    "thanks": "You're welcome!"
}

print("Chatbot is ready! Type 'bye' to stop.")

while True:
    user_input = input("You: ").strip()

    if user_input.lower() in ["bye", "goodbye", "exit"]:
        print("Bot: Goodbye!")
        break

    if not user_input:
        print("Bot: Please type something.")
```

```
        continue

    user_vector = vectorizer.transform([user_input])
    prediction = model.predict(user_vector)[0]

    print("Bot:", responses[prediction])
```

```
Chatbot is ready! Type 'bye' to stop.
Bot: Hello! How can I help you?
Bot: I am a Machine Learning chatbot.
Bot: Machine Learning helps computers learn patterns from data.
Bot: Goodbye!
```